

REVATI ENTERPRISES

FULL-STACK ARCHITECTURE, LANGUAGES, CONNECTIONS & DATA MODELS

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Generated: October 2026	Local Server: http://localhost:8080 (REST + PWA)
Target Domain: Commercial Facility & Staff Operations	Security Protocol: JWT HMAC-SHA256 & Firebase Auth

1. Executive Summary & Core Tech Stack

The **REVATI ENTERPRISES** platform is a hybrid, high-resilience facility management and enterprise workforce system. It features a responsive luxury client presentation portal, an administrative governance center, and a dedicated **STAFF APP ONLY** mobile PWA with biometric attendance, active GPS verification, and persistent duty shift timers. The system operates on a dual-tier architecture providing full offline local capability as well as scalable cloud synchronization.

Layer	Primary Language / Technology	Core Responsibilities & Components
Frontend UI	HTML5, Vanilla CSS3, JavaScript (ES6+)	Interactive client portal, Admin Dashboard, Staff Mobile App, Cost Estimator, Quotation Generator.
Backend (Local)	Python 3.12 (Multi-Threaded HTTP/1.1) <i>Alternative: Java 8+ / Spring Boot</i>	High-speed REST API server on port 8080, HMAC-SHA256 JWT auth, file & JSON document store persistence.
Cloud Backend	Google Firebase (Firestore & Auth) Vercel Edge Network	Serverless cloud document database, real-time sync listeners, Google & Email authentication, global CDN.
Networking	RESTful JSON over HTTP, CORS, WebSockets/gRPC	Cross-device communication between mobile clients, local servers, and Google Cloud endpoints.
Client Storage	LocalStorage, Cache API, Service Worker	Offline resilience, punch-state stopwatch persistence, PWA caching, zero-data-loss fallback.

2. Frontend Architecture & Languages

Languages & Standards: HTML5, CSS3, JavaScript (ECMAScript 2022+)

HTML5 Architecture: Fully semantic pages (welcome.html, index.html, services.html, estimator.html, staff.html, admin.html, register.html, letterhead.html) with complete OpenGraph metadata, PWA Manifest linking, and accessibility compliance.

Vanilla CSS3 Luxury Design System: Tailored Dark-Gold palette (--bg-main: #07090E, --gold-primary: #D4AF37), glassmorphism (backdrop-filter: blur(16px)), CSS Grid / Flexbox layouts, hardware-accelerated 3D parallax transforms, and 5 responsive breakpoints (1200px, 1024px, 768px, 480px, 375px).

Client-Side Script Modules:

- js/app.js: Main application router, modular dashboard tabs, reactive KPI counters, and data tables.
- js/api.js: JWT security client, automated multi-tier request routing (Firebase -> Java/Python API -> LocalStorage fallback).
- js/firebase-service.js: Cloud Firestore CRUD, Firebase Auth (Google OAuth & Email/Password), live snapshot subscriptions.
- js/video-intro.js: Canvas 2D cinematic particle generator, light dust rendering, and Web Audio API tone synthesis.
- js/scroll3d.js: Smooth scroll parallax effects, touch swipe gesture detection, and interactive slides.
- js/jspdf.umd.min.js & js/excelExporter.js: Client-side vector PDF generation and CSV/Excel data export engines.
- sw.js: Service Worker v4 with Network-First strategy and runtime asset caching for mobile offline access.

Icons & Visual Assets: Hybrid vector system using Multi-CDN FontAwesome 6.5.1 + embedded inline SVG symbol sprites guaranteeing 100% offline rendering without broken glyph boxes.

3. Backend Architecture & Languages

Primary Backend: Python 3.12 Multi-Threaded Daemon (server.py)

The primary backend server is written in **Python 3.12** using standard libraries (http.server, socketserver, hmac, hashlib, json, base64) to ensure zero third-party dependency vulnerabilities. It implements a multi-threaded HTTP/1.1 daemon binding to 0.0.0.0:8080.

Multi-Threading: socketserver.ThreadingMixIn with daemonized worker threads for simultaneous mobile and desktop client connections.

HMAC-SHA256 JWT Authentication Engine: Cryptographically signs and verifies JWT tokens with a 256-bit secret key, enforcing role permissions and token expiration.

Persistent Document Storage: Backed by db_store.json with atomic write locks, automated schema initialization, and data deduplication.

Full RESTful Endpoint Matrix: Provides /api/health, /api/auth/login, /api/auth/verify, /api/auth/otp/send, /api/auth/otp/verify, /api/register, /api/employees, /api/tasks, /api/attendance, /api/leaves, /api/complaints, /api/inventory, and /api/reports.

CORS & Connection Headers: Emits Access-Control-Allow-Origin: *, allows mobile devices across the local Wi-Fi subnet (e.g. 192.168.0.105:8080), and closes idle sockets cleanly with HTTP/1.1 standards.

Secondary Alternative Backend: Java 8+ / Spring Boot (CommercialHousekeepingServer.java)

A complete compiled Java backend is also provided in the repository, utilizing com.sun.net.httpserver.HttpServer with modular HTTP handlers and javax.crypto HMAC-SHA256 hashing. A Maven Spring Boot configuration (pom.xml, HousekeepingApplication.java) is structured with H2 in-memory SQL database and Hibernate JPA.

4. Connections, Protocols & Communication Channels

The system utilizes a **Multi-Tiered Fault-Tolerant Connection Topology**. Clients automatically probe cloud and local channels, falling back gracefully without crashing:

Connection Channel	Protocol / Format	Port / Transport	Security & Description
Client <-> Python API	HTTP/1.1 REST (JSON)	TCP Port 8080 (Localhost & Wi-Fi LAN)	JWT Bearer Header (HMAC-SHA256). Serves data and static HTML/CSS/JS.
Client <-> Firebase Firestore	HTTPS / gRPC / WebSockets	TCP Port 443 (Google Cloud Edge)	TLS 1.3, Firebase Security Rules, live document synchronization via onSnapshot.
Client <-> Firebase Auth	OAuth 2.0 / HTTPS	TCP Port 443	Google Identity Services popup and secure password token exchange.
Client <-> Vercel Edge	HTTP/2, HTTPS	TCP Port 443 (Global Anycast CDN)	Automated SSL/TLS, static asset delivery, and instant GitHub deployment triggers.
PWA <-> Service Worker	Service Worker Fetch API	Internal Browser IPC	Network-First caching strategy (Cache Name: revati-app-v4), offline fallback.

5. Data Models, Entities & State Architecture

All data models are designed with schema consistency across Firebase Cloud Firestore, Python JSON store, and browser LocalStorage:

Model Entity	Key Fields & Schema Attributes	Business Logic & Usage
User Entity	id, name, username, password, role, email, phone	Authentication, session tokens, and Role-Based Access Control (ADMIN, SUPERVISOR, STAFF, MANAGEMENT).
Employee / Staff	id, empCode, name, role, designation, department, status, efficiency, phone, email, proofType, proofNo, qualification, address	Complete workforce directory, biometric registration records, statutory ID tracking (Aadhaar / Passport).
Attendance Entity	id, employeeName, empCode, date, timeIn, timeOut, location, status	Biometric punch-in and punch-out records with verified GPS coordinates and shift duration tracking.
Shift Stopwatch State	isPunchedIn, inTime, startTimestamp, date	Persistent active duty timer in localStorage; prevents timer reset on mobile refresh or browser sleep.
Task Assignment	id, taskCode, title, description, location, priority, date, status ('Pending', 'Completed'), inspectedBy	Commercial cleaning schedules, facility checklists, and mobile staff verification workflows.
Leave Requisition	id, empName, empCode, leaveType, startDate, endDate, phoneOnLeave, reason, appliedDate, status, adminComment	Casual (CL), Sick (SL), and Privilege (PL) leave workflows with admin approval and emergency contacts.
Maintenance Ticket	id, ticketNo, clientName, location, priority, description, reportedDate, status, resolution	Facility issue reporting, plumbing/electrical dispatch, and resolution tracking.
Inventory Item	id, itemCode, itemName, category, quantity, reorderLevel, unit, status ('In Stock', 'Low Stock')	Industrial chemicals (Taski, EcoLab), floor scrubbers, PPE gear, and automated reorder alerts.

6. Role-Based Access Control (RBAC) Matrix

Role Name	Target User Group	Allowed Views	Key Authorizations
ADMIN	Company Directors & Operations Heads	All Views: Dashboard, Staff Directory, Leaves, Housekeeping, Inventory, Reports, Users, Letterhead	Full CRUD, register staff, delete records, approve/reject leaves, configure system settings.

SUPERVISOR	Field Area Supervisors & Leads	Dashboard, Employees, Tasks, Inventory, Reports, Attendance	Assign tasks, verify completed jobs, mark staff attendance, manage inventory levels.
STAFF	Facility Attendants, Cleaners & Techs	Dedicated Staff App Only: My Tasks, Shift Punch Clock, Leave Portal, Staff ID	Punch in/out with GPS, mark assigned duties completed, apply for leaves, view profile.
MANAGEMENT	Client Facility Representatives	Dashboard, Service Scope, Facility Reports, Complaints Desk	Log maintenance tickets, inspect site service quality, view monthly compliance reports.

TECHNICAL CERTIFICATION SUMMARY:

This architecture report confirms that Revati Enterprises utilizes an enterprise-grade, high-availability architecture comprising standards-compliant HTML5/CSS3/JavaScript frontend clients, a Python 3.12 multi-threaded REST daemon on port 8080, Google Firebase Cloud Firestore/Auth synchronization, and robust PWA service worker caching. The mobile experience features non-freezing time engines, persistent shift stopwatches, and resilient offline vector SVG iconography.